René Gateaux — Sur les fonctionnelles continues et les intégrales fonctionnelles (1919)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 17 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

Dedicated to René Gateaux, a lycée schoolmaster with no academic post, and to the hands that saved his work: his mother, who sent her dead son's papers to his school; Hadamard, who kept them safe through the war; and Lévy, who printed his corrections in small type so the editor's hand could not be mistaken for the author's.

The author's own terms

His *fonctionnelle* is a functional, his *fonction de lignes* Volterra's "function of a line." His central term is the **mean value**, printed in 1919 as a bold **M**: there being no honest finite-dimensional formula for its "integral," he takes the mean of the functional's *n*-th section over the domain's *n*-th section, then the limit as *n* grows; on the sphere of function space it is a Gaussian average. His Δ is the integral over β of the second functional derivative, and $\Delta U = 0$ is his functional Laplace equation. The "Gateaux derivative" is not his term, but the textbooks'.

The story

René Eugène Gateaux was born on 5 May 1889 at Vitry-le-François. He entered the École Normale Supérieure in 1907, passed the agrégation in 1910, and in 1912 became professor of mathematics at Bar-le-Duc.

He was preparing a thesis on the functional calculus. Borel steered him to a grant year in Rome with Volterra; arriving in late October 1913, he came home in June 1914. This memoir was, by Lévy's testimony, the last thing he wrote, dated June 1914, left among his half-finished thesis manuscripts.

Mobilised in August 1914, he headed a machine-gun section, and was killed at Rouvroy, near Lens, on 3 October 1914, aged twenty-five — the regimental date, correcting Hadamard's notice, which had said September 1914. In December 1916 the Académie des Sciences awarded the Prix Francœur to his five short notes of 1913–14; Hadamard kept the manuscripts safe through the war, and Lévy prepared them for print in winter 1918–19.

The work and what survives

The memoir, *Sur la notion d'intégrale dans le domaine fonctionnel et sur la théorie du potentiel*, appeared in the *Bulletin de la Société Mathématique de France* 47 (1919), pp. 47–70, with Hadamard's notice and Lévy's notes. Lévy published it first of the two he prepared, calling it “the richest in results,” and says the draft was not definitive in its author's mind: his modified passages in small type, important modifications footnoted and signed.

On that mechanism he builds a double-layer potential with its mean-value property, the functional Laplace equation and its Dirichlet problem, the single-layer potential, and Helmholtz-type equations. Lévy's warnings stand: one demonstration rested, in his judgement, on a faulty calculation — the result stands — and one result is inexact outside the convex case.

The reception came late. Wiener confessed in 1920 that he had not known the posthumous papers, and in 1923 named Gateaux and Lévy the deepest studies on infinite-dimensional integration; Daniell had built his own in ignorance of Gateaux, and Mazliak warns against overestimating him. Beyond it survive, untranslated in the folder, the companion memoir of 1919 (pp. 70–96), the 1922 paper, and the five notes of 1913–14.

Its English fortunes

No English translation of this memoir was found after research; its argument is expounded in English scholarship, and its results are restated wherever the Gateaux derivative is taught.

Mazliak's English study expounds its argument and translates his programmatic letter of 12 April 1913: exposition and a translated letter exist in English; a translation of the memoir itself was not found. The folder's dedication says it “has never been Englished” — too broad, and not this desk's finding. The searches were run on 16 October 2026, with no WorldCat search in this pass; the finding stands at that strength, and no stronger.

What completing it would take

The folder holds a complete draft translation of the memoir, with Hadamard's notice and Lévy's notes; what remains is verification — collation against the 1919 page images, and identification of Lévy's small-type passages, which the machine-read text cannot distinguish. Its expert gate: a working mathematician in functional analysis and a reader of the period French, before any publication. Completing Gateaux beyond it would take the companion memoir, the 1922 paper, and the five notes. The unfinished thesis cannot be completed, only printed as Lévy framed it: his text carried across, the editor's layer visible and credited, the unfinished places named.

Sources

- Project folder: `~/workspace/rare-and-obscure-books/books/gateaux-notion-integrale/gateaux-notion-integrale.txt`
- PIN: `~/workspace/packages/boneyard-extension/research/PIN-gateaux.md`
- Numdam, the 1919 printing: https://doi.org/10.24033/bsmf.994
- MacTutor, Gateaux biography: http://mathshistory.st-andrews.ac.uk/Biographies/Gateaux/
- Mazliak, “The ghosts of the Ecole Normale”: http://arxiv.org/pdf/math/0701490v2